

Nuke Code Editor 2.0

A PyCharm-style Python IDE for Foundry Nuke - PySide2 and PySide6

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Package contents	Where it goes
init.py	~/.nuke/init.py (adds the td_tools folder to Nuke's plugin path)
menu.py	~/.nuke/menu.py (TD menu, Alt+E shortcut, dockable panel)
td_tools/nuke_code_editor.py	~/.nuke/td_tools/nuke_code_editor.py (the editor, one file)
README.pdf	this document

1. Requirements

Item	Details
Nuke	13, 14, 15 (PySide2, Python 3.7 - 3.10) and 16+ (PySide6). No extra Python packages are needed.
Optional extras	jedi (smarter completion), black or autopep8 (code reformatting). The editor detects them automatically and falls back to built-in behaviour without them.
Standalone test	Outside Nuke: <code>python td_tools/nuke_code_editor.py</code> (needs PySide2 or PySide6). Node tools then report that Nuke is unavailable.

2. Installation

The Nuke user folder is **~/.nuke** (Windows: `C:\Users\<you>\.nuke`). Extract the zip into it so the layout looks like this:

```
~/.nuke/  
  init.py  
  menu.py  
  td_tools/  
    nuke_code_editor.py
```

You already have init.py / menu.py? Do not overwrite them - open the supplied files and paste their contents (everything below the docstring) at the end of yours.

Restart Nuke. You should see a new **TD** menu in the menu bar, and the Script Editor prints nothing unless something failed (look for lines starting with `[NukeCodeEditor]`).

Open the editor with **TD > Nuke Code Editor** or **Alt+E**. It can also be docked: **TD > Nuke Code Editor (panel)**, or **Windows > Custom > Nuke Code Editor**.

Alternative without `init.py` (`menu.py` only): put `nuke_code_editor.py` in any folder on `NUKE_PATH` or `PYTHONPATH`.

Tip: use the floating window for daily work. When docked inside Nuke's main window, Nuke's own shortcuts (`Ctrl+S`, `Ctrl+N`, `Ctrl+O`, `Ctrl+W` ...) can be ambiguous with the editor's.

3. Quick start

Goal	How
Run the current tab inside Nuke	Ctrl+Return (or F5 / Shift+F10). Output appears in the Run panel; each run is a single undo step.
Run only the selection / current line	Ctrl+Shift+Return
Run in a separate Python or in nuke -t	Change the combo box in the toolbar (Inside Nuke / Python (external) / Nuke -t). Set the interpreter under File > Settings.
Debug	Click the gutter (left of the line number) to set red breakpoints, then Shift+F9 .
Open anything	Tap Shift twice for Search Everywhere (actions, files, symbols, recent files).
Use a project folder	File > Open Folder as Project (Ctrl+Shift+O). The Project tree, Go to File, Find in Files and TODO scan all use it.
Insert node code	TD Tools dock > Nodes > <i>Sel -> Python rebuild</i> writes code that recreates the selected nodes and their wiring.

4. Feature overview

Area	What you get
Editor	Python highlighting (Darcula / Light), line numbers, code folding, bracket matching, auto-closing brackets and quotes, smart Enter / Backspace / Home, highlight of other uses of the word under the caret, right margin, soft wrap, show whitespace, multi-line indent, comment toggle, move / duplicate / delete / join lines, extend selection, toggle case.
Live templates	Type an abbreviation then Tab: main def defs class fori if ife try with prn prop log nsel nall nread nwrite ntn ncn nundo ncb nkc nprog. The n* templates are Nuke-specific (e.g. nsel = loop over selected nodes, nundo = undo-group boilerplate).
Inspections	Syntax errors, unresolved references, unused imports and local variables, bare except, mutable default arguments, long lines. Shown as squiggles, gutter markers and in the Problems panel. F2 / Shift+F2 jump between problems.
Completion & docs	Ctrl+Space or automatic while typing. Completes against live objects (nuke.selectedNode().), imports and document words; jedi is used when installed. F1 shows signature and docstring.
Navigation	Ctrl+B or Ctrl+Click go to a declaration (own files, project files, and the Python source of Nuke modules). Find usages (Alt+F7), Structure view, breadcrumbs, bookmarks (F11), back/forward (Ctrl+Alt+Left/Right).
Refactoring	Rename (Shift+F6, token based, current file), Surround With (Ctrl+Alt+T: if, for, try/except, with, nuke Undo group ...), Reformat (Ctrl+Alt+L: black, then autopep8, then built-in), Optimize Imports (Ctrl+Alt+O: removes unused, sorts and groups).
Search	Find / Replace bar (regex, case, whole word), Find / Replace in Files (Ctrl+Shift+F), Go to File / Symbol / Line, Recent Files (Ctrl+E), TODO panel.
Run panels	Run output, Terminal (cd supported), Python Console (interactive REPL sharing Nuke's namespace), Debug, Problems, Find, TODO.
Session	Open tabs, unsaved text, breakpoints, bookmarks, window layout and settings are restored on the next start.

5. Debugger

The debugger is built on Python's bdb and runs the script inside Nuke's own interpreter, so it sees the real Nuke objects.

Action	Key / button
Toggle breakpoint	Click the gutter, or Ctrl+F8
Start debugging	Shift+F9 or the Debug button
Resume / Step over / Step into / Step out	F9 / F8 / F7 / Shift+F8
Stop	Ctrl+F2 or the Stop button
Inspect	Debug panel: call stack (click a frame), variables, and an expression box evaluated in the selected frame

Notes. While the debugger is paused Nuke's UI is blocked (the editor keeps its own event loop running) - press Resume or Stop before working in Nuke. Stepping into library code opens that file; frames without a source file are skipped automatically. Breakpoints can be added or removed while paused.

6. TD Tools dock

Tab	Purpose
Nodes	Filterable list of all nodes (errors in red). Click selects in the DAG, double-click zooms to the node. Buttons insert toNode() calls, a names list, or full Python that rebuilds the selection including connections.
Inspector	Knob table of the selected node, auto-refreshing; optional 'only changed' filter. Double-click a knob to insert its accessor.
Snippets	15 built-in TD snippets (Reads, Writes, dependencies, callbacks, errors ...) plus your own. Insert, Run, Save as..., Delete. User snippets live in snippets.json.
Utils	Script info, class counts, missing Read files, error nodes, unused nodes (and select them), Write paths, select by class, toggle disable, reload all Reads.

7. Keyboard shortcuts

Run / Debug		Navigate	
Run	Ctrl+Return, F5, Shift+F10	Search Everywhere	Shift Shift
Run selection / line	Ctrl+Shift+Return	Go to File / Symbol	Ctrl+Shift+N / Ctrl+Alt+Shift+N
Debug	Shift+F9	Find Action	Ctrl+Shift+A
Stop	Ctrl+F2	Recent files	Ctrl+E
Resume / Over / Into / Out	F9 / F8 / F7 / Shift+F8	Go to line	Ctrl+G
Clear output	Ctrl+L	Declaration / Usages	Ctrl+B (Ctrl+Click) / Alt+F7
Edit		Back / Forward	Ctrl+Alt+Left / Right
Comment	Ctrl+/	Next / prev problem	F2 / Shift+F2
Duplicate / Delete line	Ctrl+D / Ctrl+Y	Bookmark / list	F11 / Shift+F11
Move line	Alt+Shift+Up / Down	Code	
Join lines	Ctrl+Shift+J	Complete	Ctrl+Space
Extend selection	Ctrl+W	Quick documentation	F1
Toggle case	Ctrl+Shift+U	Rename	Shift+F6
Fold / unfold all	Ctrl+Shift+- / Ctrl+Shift+=	Surround With	Ctrl+Alt+T
Find / next / prev	Ctrl+F / F3 / Shift+F3	Reformat	Ctrl+Alt+L
Replace	Ctrl+R	Optimize imports	Ctrl+Alt+O
Find in files	Ctrl+Shift+F	Settings	Ctrl+Alt+S
Font size	Ctrl+= / Ctrl+-	Terminal / Py console	Alt+F12 / Alt+F11

The full list is also available in the editor under Help > Keyboard Shortcuts (Ctrl+Shift+F1).

8. Settings and data files

File > Settings (Ctrl+Alt+S): theme, font family and size, tab width, right margin (0 disables the margin line and the long-line inspection), soft wrap, show whitespace and the external Python interpreter. All persistent data is stored in:

```
~/.nuke/NukeCodeEditor/  
  settings.json  preferences, recent files/projects, window geometry  
  session.json   open tabs, unsaved text, breakpoints, bookmarks  
  snippets.json  your saved snippets  
  tmp/           temporary copies of unsaved tabs for external runs
```

To reset the editor, close it and delete the file(s) you want to reset.

9. Troubleshooting

Problem	Fix
No TD menu	Check that menu.py is in ~/.nuke and the Script Editor shows no [NukeCodeEditor] message. Confirm the layout in section 2 (td_tools/ next to init.py).
ImportError: nuke_code_editor	init.py was not loaded or td_tools is missing. Add the folder manually: nuke.pluginAddPath('/full/path/to/td_tools').

Problem	Fix
Shortcut does nothing when docked	Use the floating window, or change the clashing Nuke shortcut.
'No interpreter found' for external run	File > Settings > Python interpreter, e.g. python3 or a full path.
nuke -t run fails	nuke -t needs a Nuke licence (render or interactive) and a Nuke executable on PATH when not run from inside Nuke.
Unresolved reference warnings that are wrong	Names created dynamically (exec, globals()) cannot be seen. Files using 'from x import *' or exec() skip that inspection automatically.
Editor looks wrong / odd state	Delete settings.json and session.json (section 8).
Slow with huge files	Inspections and occurrence highlight turn themselves off above roughly 300-500 KB of text.

10. Known limitations

- Rename and Reformat/Optimize operate on the current file; there is no project-wide refactoring.
- Rename is token based: every identifier with that name in the file is renamed (strings and comments are left alone).
- Folding is indentation based; there is no git or local-history UI, no multi-caret editing and no split view.
- Debugging pauses Nuke's UI; do not use it during renders.
- The UI code was written for PySide2 and PySide6 but is verified only where the logic is Qt-free; please report anything that misbehaves in your Nuke version together with the Script Editor output.

11. Uninstall

Delete `td_tools/nuke_code_editor.py` and remove the supplied blocks from `init.py` and `menu.py`. Optionally delete `~/nuke/NukeCodeEditor` to remove saved settings.